

Artist E-commerce: Sales Module, Purchase Module, Artwork Fraud Detection Module, Payment Module, AI Customer Service Chatbot Module

By

Au Yong Ming Jian



FACULTY OF COMPUTING AND
INFORMATION TECHNOLOGY

TUNKU ABDUL RAHMAN UNIVERSITY OF
MANAGEMENT AND TECHNOLOGY
KUALA LUMPUR

ACADEMIC YEAR
2025/26

Artist E-commerce: Sales Module, Purchase Module, Artwork Fraud Detection Module, Payment Module, AI Customer Service Chatbot Module

By

Au Yong Ming Jian

Supervisor: CHOY LAI FUN

A project report submitted to the
Faculty of Computing and Information Technology
in partial fulfillment of the requirement for the
Bachelor of Information Technology (Honours)

Faculty of Computing and Information Technology
Tunku Abdul Rahman University of Management and Technology
Kuala Lumpur

Copyright by Tunku Abdul Rahman University of Management and Technology.

All rights reserved. No part of this project documentation may be reproduced, stored in retrieval system, or transmitted in any form or by any means without prior permission of Tunku Abdul Rahman University of Management and Technology.

Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions.



Au Yong Ming Jian

Bachelor of Information Technology (Honours) in Software Systems Development

ID: 24WMR0934

Abstract

This is a **one-page** summary of the project (without any subheading) of usually not more than **300 words**. It is often used to help the reader to quickly ascertain the documentation's purpose. Like all summaries, abstract covers the main points of a piece of writing that includes the field of study, problem statement, methodology adopted, research process, conclusion and planning of the project work, etc. It helps readers understand the project by acting as a pre-reading outline of key points.

The contents that you may include in an abstract are as follows:

Purpose: Describes the main purpose of carrying the project, i.e. to justify the existence of the project by stating the existing problem to be solved. Explain why it is necessary to solve it and how the project contributes to the solution.

Scope: This covers areas or size of the project in terms of function and features of the system, modules or sub-modules involved, and functional areas covered in an organization.

Methodology: To describe the methodology, tools, techniques and models that are used throughout the project.

Testing criteria used: To describe the various assessment areas this project has undergone.

Results and Conclusion: To brief the outcome, strengths and weakness of the system or the entire project.

Note: Your abstract **SHOULD NOT** have bullet points or headings according to the contents listed above. Refer to abstracts of research papers for examples of abstracts.

Acknowledgement

This contains acknowledgement to those who have contributed directly or indirectly to the completion of the project. Usually the people to be acknowledged include the project supervisor(s), moderators, family, and those who have given assistance and supports to ensure the success of the project.

Table of Contents

1 Introduction	2
1.1 Problem Statement	2
1.2 Project Objective(s)	4
1.3 Project Background	4
1.3.1 Proposed Solution	4
1.3.2 Target Market	5
1.4 Advantages & Contribution	5
1.4.1 Tailored Towards to Visual Art Market	5
1.4.2 Tools and AI Supports	6
1.4.3 Increase Interaction in Artist Community	6
1.4.4 Contribution	6
1.5 Project Plan	7
1.5.1 Proposed Development Model	7
1.5.2 Project Scehdule	7
1.6 Project Team & Organization	8
1.6.1 Project Team	8
1.6.2 Project Scope	9
1.6.3 Project Organization	11
1.7 Chapter Summary and Evaluation	13
2 Literature Review	15
2.1 Sub-section Heading	15
2.1.1 Review of Existing System	15
2.2 Comparison of Existing System	17
2.2.1 Comparison of Sales Module	17
2.2.2 Comparison of Purchase Module	18
2.2.3 Comparison of Artwork Fraud Detection Module	19
2.2.4 Comparison of Payment Module	19
2.2.5 Comparison of AI Customer Service Chatbot Module	20
2.3 Comment	21
2.4 Proposal	22
2.5 Chapter Summary and Evaluation	24
3 Methodology and Requirements Analysis	26
3.1 Sub-section 1 Heading	26
3.1.1 Sub-subsection Heading	26
3.2 Sub-section 2 Heading	26
3.3 Chapter Summary and Evaluation	26
4 System Design	28
4.1 Sub-section 1 Heading	28
4.1.1 Sub-subsection Heading	28
4.2 Sub-section 2 Heading	28
4.3 Chapter Summary and Evaluation	28

5 Implementation and Testing	30
5.1 Sub-section 1 Heading	30
5.1.1 Sub-subsection Heading	30
5.2 Sub-section 2 Heading	30
5.3 Sub-section 1 Heading	30
5.3.1 Sub-subsection Heading	30
5.4 Sub-section 1 Heading	31
5.4.1 Sub-subsection Heading	31
5.5 Chapter Summary and Evaluation	31
6 System Deployment	33
System Backup and Risk Management	33
On-site Setup	33
Training Procedure	33
Follow-up	33
Chapter Summary and Evaluation	33
7 Discussions and Conclusion	35
Summary	35
Achievements	35
Contributions	35
Limitations and Future Improvements	35
Issues and Solutions	35
References	36

Chapter 2

Literature Review

2 Literature Review

This chapter reviews existing literature relevant to the development of an artist-focused e-commerce platform. It highlights the **key features** commonly found in existing art-centered e-commerce systems and provides a **comparative analysis of major market competitors**. The review also considers both the **advantages and limitations** of these platforms for emerging or local artists. In addition, this chapter explores the common technologies used in the development of these systems. These insights are essential in identifying best practices, addressing current system gaps, and guiding the development of a platform that is tailored to the needs of the artists.

2.1 Sub-section Heading

In the current digital world, technological advancements have helped transform and digitalize the methods of marketing and selling artwork products, such as visual art, sculpture, and drawings, to the online marketplace. Traditional methods such as selling art in galleries or at art fairs are being replaced or complemented by online platforms, as technology becomes more deeply integrated into the art market. For example, the "Art Market Malaysia" that was founded by Hazrita Mohd Hatta is an online art exhibition that helps aspiring artists present their work to the public during the COVID-19 pandemic, where physical exhibitions became unfeasible (Reena Sekaran, 2020). This transition allowed over 100 local artists, including individuals with autism and health-related challenges, to showcase their creativity to a broader audience from the comfort of their homes (Reena Sekaran, 2020).

As a result, various artist-focused e-commerce platforms have emerged, such as **Shopify**, **Etsy**, and **Squarebase**, offering a platform where artists can showcase and sell their work. These existing platforms show the growing adoption of digital tools in the art market, allowing artists to promote and sell their work through the Internet. However, each platform has its own limitations, often requiring artists to carefully evaluate and choose the one that best suits their career needs and business goals.

2.1.1 Review of Existing System

Shopify is an platform that allows artist to sell their artwork product by allowing them to build and customize their e-commerce websites. The technology Shopify uses is a **hosted SaaS (Software as a Service)** platform, where it provides tools to build an online store, add

products, accept payments, manage shipping, and track sales with a monthly subscription (Richard Nguyen, 2025). All of the server maintenance, hosting, and infrastructure are managed by Shopify itself. Shopify also lets artists use **third-party payment gateways** such as **Stripe** and **PayPal** for their e-commerce payment processes, which are fast and secure. Moreover, Shopify supports **app integration** to extend the platform's functionality, including features such as **AI-powered customer support**, which allows artists to customize their e-commerce stores to meet their specific needs.

The **characteristics of Shopify** are that this platform allows artists to sell both digital and physical artwork, enabling them to reach different markets and expand their sales model to suit their business goals. Not only that, but Shopify also offers **full control over their branding and storefront**, allowing artists to **customize their e-commerce websites** without any limitations. Shopify operates on a **subscription-based model** and **charges transaction fees monthly or yearly** based on the plan that the artist chooses, such as basic, grow, and advanced plans.

Etsy is also another e-commerce platform for artists to showcase their artwork products. Etsy transitioned from an on-premise data center to a **cloud-based infrastructure**, where it provided greater scalability and flexibility to effectively meet the demand during COVID-19 (Nicole Deslandes, 2024). Not only that, Etsy also uses **Artificial Intelligence (AI)** and **Machine Learning (ML)** to enhance the Etsy e-commerce functionality. Etsy uses AI to build a **recommendation system** that recommend artwork products to customers who might be interested in them based on their browsing history and preferences (Nicole Deslandes, 2024). In addition, Etsy also uses AI to build a **chatbot** for its **customer service** to help resolve customer delivery issues, shop listing issues, refund issues, and account issues (Etsy, 2025). Etsy also uses ML to build a system that **flags artwork products** that violate Etsy's policies to improve trust and safety (Nicole Deslandes, 2024).

The **characteristics of Etsy** are that the platform supports a mainly physical artwork, such as **DIY handcrafts, physical drawings, and art prints**, making it a suitable marketplace for independent artists to reach a wider audience base. Additionally, Etsy has a **user-friendly interface** that provides a **sales dashboard, inventory management, order tracking, and shipping tools**, allowing users to easily manage their storefronts, create product listings, and customize shipping methods, making it accessible even for those with minimal technical

skills (Abas Khan, 2025). Moreover, Etsy charges artists based on a **pricing structure** that includes **several types of fees**, such as listing fees, transaction fees, payment processing fees, and shipping fees.

Squarebase has a similar concept to **Shopify**, but it is very different from it. Squarebase is a **website builder** that allows artist to build their artist e-commerce using their provided templates. Same as Shopify, Squarebase runs as a hosted **SaaS platform**, which allows artists to build and manage their e-commerce websites within the **Squarebase suite** without requiring coding knowledge (SPSP Themes, 2024). Squarespace also supports integrated third-party payment gateways that allow artists to accept various payment options through services such as **Stripe**, **PayPal**, and **Apple Pay**. It also has other third-party tools to integrate into artist e-commerce websites, such as **Google Analytics**, **MailChimp**, and **social media** like **Instagram**, to enhance their e-commerce functionality (SPSP Themes, 2024).

The **characteristics of Squarebase** are that it has **drag-and-drop features** to move things like **images, buttons, and text** to any spot that the artist wants (Druv Nimbark, 2025). Combined with a selection of designed templates, this feature enables artists to create visually appealing and personalized e-commerce websites with minimal technical effort. It also supports **physical and digital artwork** for the artist to sell on their platform. Furthermore, it operates on a **subscription-based model**, requiring artists to pay a monthly fee in exchange for access to its integrated tools, hosting services, and e-commerce functionalities (SPSP Themes, 2024).

2.2 Comparison of Existing System

This section presents a comparison of existing platforms based on key features relevant to visual artists, such as **sales module, purchase module, artwork fraud detection module, payment module, and AI customer service chatbot module**. By comparing these features, we can better understand the **advantages and limitations** of each platform in supporting the needs of creative professionals. The comparison focuses on three major platforms, which are **Shopify, Etsy, and Squarespace**.

2.2.1 Comparison of Sales Module

Table 2.1 Comparison of Sales Module

Features	Shopify	Etsy	Squarespace
----------	---------	------	-------------

Supported Product Types	Physical and digital artwork	Physical artwork	Physical and digital artwork
Inventory Management	built-in tools or third-party tools	Only built-in tools	built-in tools
Storefront Design Control	User have full control over branding	Etsy branding dominates	Full design control
Analytics and Reporting	built-in tools or third-party tools	built-in tools	built-in tools or third-party tools
Advantages	1) Supports physical and digital artwork 2) Complete control over branding 3) A lot of supported integration	1) No upfront monthly cost 2) Easy for an emerging artist to use 3) Ideal for reaching a wider audience without marketing	1) Supports physical and digital artwork 2) Lower commission 3) Offers full design control
Disadvantages	1) Complex charges could be more costly 2) Monthly fees are expensive	1) Hard to stand out due to a crowded marketplace 2) Limited design control 3) Only support physical product	1) Marketplace exposure is limited 2) Fewer third-party integrations 3) Customization can be restrictive if you want fancy features

2.2.2 Comparison of Purchase Module

Table 2.2 Comparison of Purchase Module

Features	Shopify	Etsy	Squarespace
Recommendation System Availability	Built-in and third-party integration	Built-in	Limited third-party integration
Cart Management	Advanced cart functionality	Basic cart functionality	Basic cart functionality
Product Rate & Review	Require third-party integration	Built-in	Require third-party integration
Customizable Checkout	Highly customization	No customization	Limited customization
Search & Filters Availability	Advanced search with filters	Basic built-in search with filters	Basic built-in search with filters

Advantages	1) Full control over checkout customization 2) Offer advanced features	1) User-friendly for customers 2) No extra cost for the features	1) User-friendly for customers 2) No extra cost for the features
Disadvantages	1) Costly for advanced features 2) Complex to integrate third-party tools	1) No checkout customization 2) No control over the recommendation system	1) Minimal checkout customizations 2) Limited recommendation features

2.2.3 Comparison of Artwork Fraud Detection Module

Table 2.3 Comparison of Artwork Fraud Detection Module

Features	Shopify	Etsy	Squarespace
Availability	Third-party integration	Built-in	Not available
AI-powered Detection	Available via third-party integration	Available	Not available
Content Moderation	Merchant responsibility	Manually reviews reports & suspends violators	Merchant responsibility
Advantages	1) Allows integration of third-party tools for fraud detection 2) Flexible to choose a fraud detection app	1) Actively monitors content via community flags 2) Has built-in fraud detection mechanisms	1) Full content control and site ownership
Disadvantages	1) Extra cost and technical knowledge required to integrate	1) Manual review can lead to delayed action 2) No full control of the AI-powered detection	1) No fraud detection available 2) Relies on user report

2.2.4 Comparison of Payment Module

Table 2.4 Comparison of Payment Module

Features	Shopify	Etsy	Squarespace
Availability	Built-in or third-party	Built-in	Built-in or third-party

Third-Party Payment Gateway Supported	Stripe, Paypal, ApplePay	Only Etsy Payment	Stripe and Paypal
Transaction fee	Charges are based on the vendor and plan chosen by the user	Charge RM0.85 per listing + 6.5% per sales	Charge 3% per sales on Business plans but 0% on Basic and Advanced Plans.
Ease of use	Moderate to use	Simple to use	Simple to use
Advantages	1) Support built-in and third-party 2) Flexible payout options	1) Simple and clean interface 2) Automatic tax and currency handling	1) Support built-in and third-party 2) Simple and clean interface
Disadvantages	1) Transaction fees apply if not using Shopify Payments 2) Setup can be complex	1) High transaction fee 2) Limited control over the payout process	1) Limited Payment Gateway Options 2) No proprietary payment system

2.2.5 Comparison of AI Customer Service Chatbot Module

Table 2.5 Comparison of AI Customer Service Chatbot Module

Features	Shopify	Etsy	Squarespace
Availability	Third-Party apps	Built-in	Built-in
Customization by Artist	Full control to customize the chatbot	Not customizable	Not customizable
Ease of Use	Moderate to use	Simple to use	Simple to use
Order Tracking	Yes, if it's been configured in the third-party app	Built-in	Not supported
Advantages	1) Supports powerful chatbot integrations 2) Full control for artist branding, FAQs, and workflows	1) Built-in chatbot with no setup required 2) Simple for the customer to use	1) Built-in chatbot with no setup required. 2) Simple for the customer to use

Disadvantages	1) Requires manual setup of third-party apps 2) May involve additional cost	1) Not customizable by the artist 2) Limited to Basic customer support queries	1) Not customizable by the artist 2) Primary assists the artist only
---------------	--	---	---

2.3 Comment

Refer to Table 2.1 for the comparison of the **Sales module**, which shows that **Shopify** sales module offers the most flexibility for artists. This is because the artist has **full control of their storefront**, allowing them to customize their e-commerce platform freely, although it requires a higher cost to get those features. **Etsy**, while easy to use and ideal for reaching a wide audience, is limited in terms of customization and only supports physical artwork. **Squarespace** provides a balanced approach with design freedom and lower commissions but lacks exposure and a third-party app ecosystem, which may limit advanced customization.

Refer to Table 2.2 for the comparison of the **Purchase module**. The result shows that **Shopify** stands out with its advanced cart, search, and customizable checkout experiences. The reason is that Shopify allows the artist to have **full control of the checkout customization**, enabling them to tailor the purchase flow to match their branding. It also offers a powerful recommendation system that supports both built-in and third-party integration to provide personalized shopping experiences for its customers. However, these features are costly and complex to implement due to the requirement to integrate with third-party applications. **Etsy** offers a user-friendly and low-barrier solution with built-in recommendation and review features, but lacks flexibility in checkout design. **Squarespace** shares a similar ease-of-use with Etsy but lacks in recommendation systems and checkout customization, making it more suitable for smaller businesses.

Refer to Table 2.3 for the comparison of the **Artwork Fraud Detection module**, which shows **Etsy** has a better solution compared to **Shopify and Squarespace**. The reason is that **Etsy** provides built-in fraud detection mechanisms supported by community flags reporting. This allows Etsy to actively monitor and address suspicious listings, offering better protection for artists against content misuse and duplication. The downside of it is that the artist has no control over the AI-powered detection system, which may lead to confusion when the product is flagged as suspicious, but it is an actual original artwork. **Shopify**, while lacking a native

solution, allows integration of third-party fraud detection tools, giving artists full control at the cost of added complexity and expense. **Squarespace**, on the other hand, offers no built-in fraud detection and relies on user reporting, making it less ideal for protecting artwork from duplication or misuse.

Refer to Table 2.4 for the comparison of the **Payment module**. It shows that **Shopify** has more **payment gateway** options compared with **Etsy** and **Squarespace**. This is because **Shopify** supports third-party payment gateways, offering flexibility and broad options for artists. However, the setup of the third-party payment gateway can be **complex**, which may cost some time to configure. **Etsy** only has a built-in payment system that simplifies the payment process, which gives artists less control. Additionally, it imposes high transaction and listing fees, which can affect the profitability of the artist. Although **Squarespace** supports third-party payment gateways, the options are limited to PayPal and Stripe and also lack a proprietary payment solution, which may restrict flexibility for artists seeking a broader range of payment solutions.

Refer to Table 2.5 for the comparison of the **AI Customer Service Chatbot module**. It shows that **Shopify** is in the lead with its customizable third-party chatbot apps that enable artists to manage FAQs, orders, and branding via AI-powered tools, but these must be installed and could have additional charges. Both **Squarespace** and **Etsy** provide basic, built-in chatbot software that is less customizable and functional but more user-friendly, and thus sufficient for basic support but not powerful enough for more complex customer service needs.

2.4 Proposal

Based on the analysis across the modules, which are **Sales, Purchase, Artwork Fraud Detection, Payment, and AI Customer Service Chatbot modules**, it is clear that no single existing platform fully addresses the needs of the artists and consumers. Therefore, the proposed artist-focused e-commerce platform will be designed by incorporating the strengths and addressing the limitations identified in each comparison.

In order to provide flexibility and accommodate different artistic practices, the suggested system will **accommodate both physical and digital artwork**. Inspired by Etsy's easy-to-use interface, the suggested platform will value **simplicity and ease of use**. It will provide an easy setup process and integrated tools that enable artists to list their artwork easily without needing technical expertise.

The shopping experience will include a user-friendly interface with **cart functionality, product rating & review system, and search & filter**. A crucial feature will be the **recommendation system** that will be personalized, with the aim to allow users to discover artwork that is relevant to their **browsing and purchasing behaviour**. Inspired by Shopify's flexibility and Etsy's usability, the system will have a recommendation engine that will offer relevant product recommendations to every user while promising a seamless and efficient shopping experience for different classes of users.

The platform will implement built-in **AI-powered artwork fraud detection**, improving upon community moderation by **combining automated scanning with manual flagging**. Unlike Squarespace, which lacks any protection, the proposed system will provide artists with tools to report and track potential copyright violations, as well as watermark options to reduce theft risk.

The proposed system will utilize **Stripe** as the **default third-party payment gateway**. Stripe offers secure, trusted transactions and accepts a wide range of payments, including credit and debit cards. The system will also provide artists with **flexible payout terms and transparent fees**, addressing the cost problem prevalent in sites like Etsy.

The proposed system will also feature a built-in **AI chatbot** to **handle basic customer support queries and order tracking**, similar to the approach used by Etsy and Squarespace. The chatbot will come pre-trained with standard response templates and automated workflows, ensuring a seamless support experience without requiring artists to configure or

customize it. This reduces the setup burden for artists while maintaining efficient customer communication.

2.5 Chapter Summary and Evaluation

In conclusion, this chapter reviews and compares the technology and features used by the existing artist-focused e-commerce system. The analysis reveals that while current e-commerce systems offer useful capabilities, none fully meet the specific needs of both artists and consumers. After reviewing the advantages, limitations, and best practices of these platforms, it becomes evident that a new system is necessary, one that is purpose-built to address the unique requirements of independent artists. The findings in this literature review have been made available as the basis for the proposed system design. It guides the development of a more artist-focused e-commerce that not only serves the business aspect of selling art but also offers community involvement, brand value, and freedom of art for artists within the online marketplace.

References

- Adnan, A. M., Abdul Manap, N., & Zakaria, Z. (2023). Increase in online purchase fraud cases: Business Ethics vs Consumers' attitudes. *International Journal of Academic Research in Business and Social Sciences*, 13(6). <https://doi.org/10.6007/ijarbss/v13-i6/16809>
- Deslandes, N. (2024, December 4). How etsy's technology strategy drives innovation & enhances UX. TechInformed. <https://techinformed.com/etsy-technology-strategy-ai-innovation-cx-ux/>
- Entrepreneur. <https://www.entrepreneur.com/starting-a-business/why-niche-marketplaces-are-the-new-wave-of-ecommerce/477166>
- Etsy. (2025). How does etsy's customer support chatbot work? – etsy help. Etsy Help Center. <https://help.etsy.com/hc/en-us/articles/27283630080151-How-Does-Etsy-s-Customer-Support-Chatbot-Work>
- Khairudin, K. A., Ali Yasin, S. M., Abdul Jamil, S. E., & Yahaya, M. H. (2023). Exploring a new market for painting in Malaysia: A need assessment analysis. *International Journal of Academic Research in Progressive Education and Development*, 12(2). <https://doi.org/10.6007/ijarped/v12-i2/17150>
- Khan, A. (2025, June 11). Pros and cons of Etsy no one talks about!. nanochipinc.com. https://nanochipinc.com/pros-and-cons-of-etsy-no-one-talks-about/?utm_source=chatgpt.com
- Liong, M.-Y., Chin, Y.-M., & Chang, Y.-F. (2024). Visual art and craft consumption pattern: An empirical analysis of Malaysian Middle-Class art consumer. *Advanced International Journal of Business, Entrepreneurship and SMEs*, 6(21), 16–34. <https://doi.org/10.35631/aijbes.621002>
- Malaysiakini. (2021, November 23). Anticipating 2022 for the Creative Industry. <https://www.malaysiakini.com/advertorial/600310>
- Nguyen, R. (2025, March 28). How does shopify work? what most new sellers get wrong. PageFly. <https://pagefly.io/blogs/shopify/how-does-shopify-work>
- Nimbark, D. (2025, July 9). What is Squarespace? pros, cons, and key features explained. Yudha Global. <https://www.wixseoexpert.com/post/squarespace-pros-cons-and-key-features-explained>
- Raeburn, A. (2025, February 13). What is Extreme Programming (XP)? [2025]. Asana. <https://asana.com/resources/extreme-programming-xp>
- Raj, A. (2022, November 2). E-commerce checkout in Malaysia is filled with errors. TechWire Asia. <https://techwireasia.com/2022/10/e-commerce-checkout-in-malaysia-are-filled-with-errors/>
- Bogdan, S. (2024, July 26). Why niche marketplaces are the new wave of Ecommerce.
- Sekaran, R. (2020, October 25). Online art market helps aspiring artists sell their work. Free Malaysia Today | FMT. <https://www.freemalaysiatoday.com/category/highlight/2020/10/26/online-art-market-helps-aspiring-artists-sell-their-work>
- Shi, P. S., & Fahim, N. A. (2024). Factors influencing customer satisfaction in shopee Malaysia. *International Journal of Academic Research in Business and Social Sciences*, 14(12). <https://doi.org/10.6007/ijarbss/v14-i12/24196>
- Themes, S. (2024, August 11). What framework is Squarespace built on?. SQSP Themes. <https://www.sqsptthemes.com/squarespace-faqs/what-framework-is-squarespace-built-on?srsId=AfmBOooZN4G29sRy9MNEyTe0mCdqLAivG6AbHR6sBQMjom6H3qz5r-wy>